GCAP3123 Computer Technology and AI Project Project 2026
Progress Log-book

Project Information

Project Title: Region-Controlled Style Transfer via Two-Stage VGG and Diffusion Integration

Group number: Group 7

Team Members:

1. Zhengda SI, 2230024178, AI;
2. Yiyi LIN, 2230034031, AI;
3. Xinyu ZHANG, 2230034061, AI;
4. Kai WEN, 2230034054, AI;
5. Langqi ZHONG, 2230034066, AI;

Project Area: Computer Vision and Artificial Intelligence Generated Content

Supervisor: Monica

Weekly Log Entry (You need to complete this log book 5 times for week 5/6/7/8/9 and submit in week 5/7/9)

Week 5

Date: 28 March 2026
Meeting Mode: □ In-person □ Online □ Mixed

1. Work completed this week (use bullet points)

- A pre-trained VGG-19 model is used to construct a feature extractor to extract content and style features from the input image.
- The input image processing module has been completed, including image loading, format unification, resizing, and tensor manipulation.
- Style and content losses based on the Gram matrix were constructed for style transfer optimization.
- Perform the complete neural style transfer optimization process and test the model performance.

2. Objectives for this week (challenge and solutions)

Challenge:

- Implement an efficient neural style transfer algorithm to ensure real-time performance when processing high-resolution images.
- Optimize model inference speed and reduce video memory (VRAM) usage.

Solutions:

- Use QThread to decouple style transfer computation from the main thread.
- Update the progress bar and display results using the signal-slot mechanism.
- Provide parameter input controls (e.g., number of iterations, style/content weights) and instant save functionality to enhance usability.
- Freeze VGG19 model parameters by setting requires_grad=Falseto to reduce unnecessary computational overhead.
- Use the torch.no_grad()context manager during feature extraction to disable gradient tracking, thereby accelerating inference.

3. Methods & Tools used

Data / Literature / Models:

Data: input image from user

Literature: L. A. Gatys, A. S. Ecker and M. Bethge, "A neural algo-rithm of artistic style," Cornell University Library, arXiv.org,Ithaca, 2015.

Models: models.vgg19(pretrained=True)

Software / Tools:

Tools: PyTorch, PyQt5, Pillow (PIL.Image), torchvision.transforms

4. Meeting with Supervisor? □ Yes □ No

Date & Time:

Feedback:

5. Individual Contribution (only list the one had contribution)

Zhengda SI: Responsible for building the feature extraction component of the model using the pre-trained models.vgg19 model. Implement the selection of content and style layers and complete the forward propagation structure of the model.

Yiyi LIN: Build the style loss calculation method based on the Gram matrix, while considering computational efficiency to reduce video memory (VRAM) usage and optimize inference speed. Build the content loss calculation method and implement the weighted fusion of the total loss (style loss + content loss).

Xinyu ZHANG: Complete the image input module, implementing loading from file paths or PIL image objects. Responsible for the complete pipeline of image format unification, size adjustment (e.g., Resize), and conversion to PyTorch tensors.

Kai WEN: Implement the complete neural style transfer optimization process, including setting up the optimizer (Adam) and writing the iteration loop. Integrate the complete chain of forward propagation, loss calculation, backward propagation, and parameter updates within the loop.

Langqi ZHONG: Integrate the above modules into a unified NeuralStyleTransfer class and ensure it is callable. Perform preliminary model testing and performance verification, recording the output results.

Week 6

Date: 3 April 2026

Meeting Mode: □ In-person □ Online □ Mixed

1. Work completed this week (use bullet points)

- Continued testing the original VGG-based neural style transfer pipeline and found that the results still lacked fine detail and visual richness.
- Studied diffusion-based image generation methods and evaluated whether Stable Diffusion Img2Img could serve as a refinement stage for the project.
- Investigated the environment requirements for diffusion deployment, including package compatibility, model loading, and local cache configuration.
- Repeatedly tested and debugged the integration of the diffusion module into the existing system.
- Successfully integrated the diffusion module after several attempts, and the system now runs with diffusion-based refinement.

2. Objectives for this week (challenge and solutions)

Challenge:

- The original VGG-based style transfer pipeline worked, but the generated images still lacked sufficient detail enhancement and visual impact.
- At the beginning of the week, we were unsure which technical direction would best improve the current output quality.
- Integrating a diffusion model into the existing project required us to resolve practical issues in environment setup, model loading, and system compatibility.

Solutions:

- Evaluated the limitations of the current VGG-based method through repeated testing and identified image refinement as the main improvement target.
- Explored possible improvement directions and selected Stable Diffusion Img2Img as the most suitable approach at this stage.
- Adjusted the environment settings and repeatedly tested model loading until the diffusion module deployed successfully.
- Connected the diffusion model to the current pipeline and confirmed that the system could run end to end successfully.

3. Methods & Tools used

Data / Literature / Models:

1. Data:
    1) Content images collected from the BNBU official website.
    2) Style reference images collected from online classical painting resources.

2. Literature:
    1) L. A. Gatys, A. S. Ecker and M. Bethge, “A Neural Algorithm of Artistic Style,” arXiv, 2015.
    2) Stable Diffusion Img2Img related online documentation and pretrained model resources.

3. Models:
    1) VGG-based Neural Style Transfer model (custom implementation)
    2) runwayml/stable-diffusion-v1-5 for diffusion-based image refinement.

Software / Tools:

1. PyTorch
2. Pillow (PIL)
3. PyQt5
4. Diffusers
5. Hugging Face Hub
6. torchvision.transforms

4. Meeting with Supervisor? ☑ Yes □ No

Date & Time: 2 April 2026

Feedback:

During last week's consultation, we got a lot of helpful advice, as follows:

A. Technical Validation & Experimental Comparison

Baseline Comparison: We must clearly contrast the outputs of "Pure VGG generation," "Pure Stable Diffusion (SD) generation," and our "Two-stage hybrid workflow." The objective is to highlight our model’s unique advantages in Structure Preservation and Semantic Detail.

Performance Metrics: Evaluation cannot rely solely on subjective observation. We need to introduce quantitative metrics, such as convergence analysis of Content/Style Loss or user study data.

Limitation Analysis: The teacher explicitly requested the inclusion of "Bad Cases." We are required to identify specific scenarios where the model fails and provide a technical post-mortem on the underlying causes.

B. Market Value & Social Impact

Real-world Demand Discovery: The project must transcend being a "toy" application. We need to investigate the practical utility of this technology in industries such as fashion design (e.g., reducing design cycles), architectural visualization, or digital marketing.

C. Core principles of the model

Mathematical Foundation: The final report must include formal mathematical descriptions of the core algorithms(Formulas are provided, as well as detailed explanations). For example, the Style Loss function.

Literature Review: Multiple academic papers need to be cited. Analyze the existing related models and their shortcomings, so as to better explain the necessity and innovation of our improved model.

5. Individual Contribution (only list the one had contribution)

Zhengda SI: Evaluated VGG-19 limitations and curated the core dataset of BNBU campus images and style references for baseline testing.

Yiyi LIN: Researched Stable Diffusion Img2Img integration and designed the "VGG-Foundation + Diffusion-Refinement" hybrid workflow.

Xinyu ZHANG: Configured the PyTorch/Diffusers environment and optimized backend model loading and memory management for local deployment.

Kai WEN: Integrated VGG and Diffusion modules into a unified pipeline and conducted end-to-end debugging to ensure operational stability.

Langqi ZHONG: Developed the image preprocessing pipeline and optimized the PyQt5 interface for better parameter control and asset management.

Week 7

Date: 10 April 2026
Meeting Mode: □ In-person □ Online □ Mixed

1. Work completed this week (use bullet points)

- Completed integration of Gradio web interface for the AI image style transfer system
- Built a unified UI combining: 1. VGG-based Neural Style Transfer (NST) 2. Stable Diffusion Img2Img refinement pipeline
- Implemented dual-mode workflow: 1. NST → Diffusion refinement pipeline 2. Diffusion-only Img2Img mode
- Added preset prompt system (art styles, cyberpunk, anime, etc.)
- Added scene and color tone control modules for better controllability
- Implemented dynamic UI interaction (e.g., enabling custom prompt textbox)
- Integrated progress callback system for real-time status updates
- Connected backend models: 1. NeuralStyleTransfer (VGG-based model) 2. DiffusionRefiner (Stable Diffusion refinement module)
- Tested full pipeline end-to-end with image input/output consistency

2. Objectives for this week (challenge and solutions)

1. **Objective 1: Integrate NST + Diffusion into a unified web demo**
   Challenge: NST output (tensor) and Diffusion input (PIL image) format mismatch
   Solution: Implemented tensor_to_image() conversion pipeline
2. **Objective 2: Ensure smooth Gradio interaction and usability**
   Challenge: UI components needed dynamic enabling/disabling (custom prompts, diffusion toggle)
   Solution: Used gr.Textbox(interactive=...) and .change() event callbacks
3. **Objective 3: Maintain stable progress feedback during long inference**
   Challenge: NST + diffusion have different execution time distributions
   Solution: Designed weighted progress system (NST 90% + Diffusion 10%)
4. **Objective 4: Improve controllability of generation results**
   Challenge: Single prompt was not expressive enough
   Solution: Added structured controls (scene preset, color tone preset, prompt presets)

3. Methods & Tools used

Data / Literature / Models:

- The two models/pipelines implemented in the previous steps:
  1. VGG-based Neural Style Transfer model (custom implementation)

2. Stable Diffusion Img2Img pipeline (via DiffusionRefiner)

- Predefined prompt engineering templates for artistic control
- Images for test and demo:
  1. Style images: 3 images for each style
  2. Content images: Real scene pictures taken by us / Images collected from publicly available online sources (e.g., Google Images, Pinterest, Unsplash)
- Literature:
  1. Abid, Abubakar, et al. "Gradio: Hassle-free sharing and testing of ml models in the wild." *arXiv preprint arXiv:1906.02569* (2019).
  2. Radhakrishnan, Unnikrishnan. "Documenting SME processes with conversational AI: from tacit knowledge to BPMN." *IET Conference Proceedings CP942*. Vol. 2025. No. 28. Stevenage, UK: The Institution of Engineering and Technology, 2025.

Software / Tools:

- How to build a **Gradio** website: https://huggingface.co/docs/hub/en/spaces-sdks-gradio
- PyTorch (tensor computation and model execution)
- torchvision (image preprocessing and transformation)
- PIL (image handling and conversion)
- Custom Python modules (completed in the previous steps):
  1. diffusionPro.py
  2. VGGmodel.py

4. Meeting with Supervisor? □ **Yes** □ **No**

Date & Time:

Feedback:

5. Individual Contribution (only list the one had contribution)

Zhengda SI: Constructed the overall frame layout of the Gradio web interface, establishing a unified UI that integrates the VGG-style transfer and Stable Diffusion refinement pipelines. This involved the structural implementation of the dual-tab system (Hybrid Mode and SD-only Mode) to provide a cohesive foundation for the system's functional modules.

Yiyi LIN: Implemented the dual-mode workflow logic to ensure seamless connectivity between the backend modules (VGGmodel.py and diffusionPro.py). Optimized the "NST → Diffusion" pipeline to manage automated data flow from the initial stylistic mapping to the final refinement stage.

Xinyu ZHANG: Engineered the dynamic interaction system using event callbacks to enable responsive components, such as custom prompt textboxes. Designed a weighted progress callback system (90% NST + 10% Diffusion) to provide accurate real-time status updates during inference.

Kai WEN: Developed structured control modules, including scene presets (e.g., cyberpunk, anime) and color tone adjustment systems. Engineered predefined prompt templates to significantly enhance the controllability and artistic expressiveness of the generated outputs.

Langqi ZHONG: Resolved critical data format mismatches through the implementation of a tensor_to_image() conversion pipeline. Conducted comprehensive end-to-end testing using a dataset of campus imagery and artistic references to ensure cross-module consistency and system stability.

Week 8

Date: 19 April 2026
Meeting Mode: □ In-person □ Online □ Mixed

1. Work completed this week (use bullet points)

- Numerous experiments were carried out with three generation modes: pure VGG style transfer, pure stable diffusion, and the proposed hybrid VGG+ diffusion.
- Results of different content images and style reference images were collected and organized for comparison.
- The three generation modes are compared from the aspects of structure preservation, visual richness, controllability, and stability.
- Start collecting the comparison cases for the final report, and particularly the cases where the hybrid method performs better than the pure VGG and pure SD methods.
- The application value of the system in clothing design, digital art generation, and advertising image production was evaluated.

2. Objectives for this week (challenge and solutions)

Challenge:

- To verify the superiority of the VGG + diffusion hybrid system, it needs to show better test results between using only VGG, using only SD, and using VGG + diffusion.
- There are still some potential issues with the outputs (structure distortion and over stylization).
- The report should have more evidence for evaluation rather than only visual inspection.

Solutions:

- Images with the same content and style were tested with three generation modes.
- We compared the outcomes according to structure, quality, controllability, and stability.
- The bad results were documented, the relevant parameters were modified, and more tests were done.
- The preliminary evaluation data for the final report has been produced.

3. Methods & Tools used

Methods：

1. Two-stage Style Transfer

* Stage 1 (Drafting): Uses VGG style transfer to create a structurally faithful, stylized base image.

* Stage 2 (Refinement): Enhances the base image via Stable Diffusion (Img2Img) guided by text prompts for artistic detail.

2. Control Mechanisms

* Enhanced Prompts: Combines user input with scene/color presets for semantic control.

* ROI Guidance: Softly blends the style image onto estimated clothing areas to localize stylization and protect other regions.

3. Evaluation

Comparative analysis of three modes:

* VGG-NST: Baseline for structure preservation.

* Stable Diffusion: Baseline for artistic quality.

* Hybrid (VGG+SD): Evaluates the combined system.

Tools：

1. Core Frameworks & Libraries:

* PyTorch: Used for building and running the VGG style transfer model.

* Hugging Face Diffusers: Used for loading and inferring the Stable Diffusion image-to-image pipeline.

* PIL / OpenCV: Used for image processing, ROI operations, etc.

2. User Interfaces:

* Gradio: Used to build an interactive web application interface for quick demonstration and testing.

* PyQt5: Used to develop a desktop application, providing stable offline experience through multi-threaded management.

3. Development & Runtime Environment:

* Python: The primary programming language.

* Hugging Face Hub: Serves as the model source. The project configures a local cache path via environment variables for storage management.

4. Meeting with Supervisor? □ **Yes** □ **No**

Date & Time:

Feedback:

5. Individual Contribution (only list the one had contribution)

Zhengda SI: Designed and executed comparative experiments for this week. Ensured that the three modes (Pure VGG, Pure SD, Hybrid) were tested on the same set of images, laying the groundwork for a fair comparison.

Yiyi LIN: Handled the technical implementation and parameter tuning for the three generation modes. Focused on ensuring the seamless handoff between the two models in the Hybrid Mode and adjusted key parameters to achieve optimal output quality.

Xinyu ZHANG: Monitored the experimental progress and organized the results. Documented the process, arranged the outputs from the three modes side-by-side, and began compiling a comparative case library for the final report.

Kai WEN: Analyzed the experimental results and summarized the key advantages. Evaluated the output quality across multiple dimensions, specifically identifying and documenting typical cases where the Hybrid Method performed better, serving as core evidence.

Langqi ZHONG: Troubleshot output issues and analyzed application scenarios. Debugged and optimized defects in the generated images and conducted a preliminary assessment of the system's practical value in fields such as clothing design, digital art, and advertising.

Week 9

Date: 25 April 2026
Meeting Mode: □ In-person □ Online □ Mixed

1. Work completed this week (use bullet points)

- Continued organizing the existing experimental results and preparing comparison materials for the final report, with a focus on the outputs generated by the VGG-based style transfer pipeline, Stable Diffusion refinement, and the hybrid VGG + diffusion workflow.
- Conducted a preliminary analysis of the current experimental results and found that there was still room for improvement in local region control, so we decided to focus on more precise garment localization and editing as the next improvement direction.
- Reviewed the current ROI-based image refinement method and found that its garment region selection was still not accurate enough, as it could be affected by factors such as human pose, clothing shape, and image background.
- Further explored feasible approaches for achieving more accurate garment localization in portrait images and finally identified human parsing/garment parsing as the main direction for subsequent improvement, providing a methodological foundation for garment mask extraction and local editing.

2. Objectives for this week (challenge and solutions)

Challenge:

- The current ROI-based local editing approach lacks precision, leading to inaccurate garment region selection and suboptimal refinement results.
- Variations in pose, clothing structure, and background make it difficult to consistently isolate garment regions using simple geometric heuristics.

Solutions:

- Introduce human parsing/garment parsing techniques to obtain pixel-level semantic segmentation of clothing regions.
- Replace rectangular ROI with mask-guided region control for more accurate and robust garment editing.
- Explore integrating mask-based editing into the existing diffusion refinement pipeline.
- Investigate auxiliary refinement methods (e.g., SAM-based mask enhancement) to improve boundary quality.

3. Methods & Tools used

Data / Models:

- VGG and Diffusion Model implemented in the previous week
- Garment-aware human parsing for semantic garment localization (`label_map`, `upper_mask`, `skirt_mask`)
- SAM 2-based mask refinement as an auxiliary boundary-cleaning method
- Mask-guided garment-region control based on extracted garment masks
- Further expandable routes considered for later stages: garment-only inpainting, ControlNet, IP-Adapter / fashion LoRA, and VTON
- Data: publicly available online images used for testing

Literature:

- Web for FASHN VTON: FASHN VTON v1.5 | Virtual Try-On Research | FASHN AI (The paper has not been published yet.)
- Ravi, N., Gabeur, V., Hu, Y. T., Hu, R., Ryali, C., Ma, T., ... & Feichtenhofer, C. (2024). Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*.

Software / Tools:

- PyTorch (tensor computation and model execution)
- torchvision (image preprocessing, transforms, and pretrained VGG features)
- diffusers (Stable Diffusion Img2Img pipeline) (Hugging Face)
- Hugging Face Hub (model loading and cache management) (Hugging Face)
- PIL / Pillow (image loading, resizing, and conversion)
- NumPy (mask and array processing)
- PyQt5 (desktop user interface)
- SAM 2 refinement script
- Garment parsing/mask extraction scripts

4. Meeting with Supervisor? □ **Yes** □ **No**

Date & Time:

Feedback:

5. Individual Contribution (only list the one had contribution)

Zhengda SI: Organized the existing experimental results for this week. Reviewed the outputs generated by the VGG-based style transfer pipeline, Stable Diffusion refinement, and the hybrid VGG + diffusion workflow, providing comparison materials for the final report.

Yiyi LIN: Collected and reviewed possible methods for garment localization. Focused on investigating human parsing/garment parsing as a potential solution for more accurate garment region detection and future mask-based local editing.

Xinyu ZHANG: Reviewed the current ROI-based refinement method and analyzed its limitations. Examined why the existing ROI selection was not accurate enough for garment-specific editing, especially under variations in pose, clothing shape, and image background.

Kai WEN: Helped narrow down the next improvement direction to garment-region editing. Compared possible local editing directions and summarized why garment areas should be treated as a more specific and controllable target region.

Langqi ZHONG: Conducted a preliminary analysis of the current experimental results and identified the main limitations of the system. Focused on locating the problems in local region control and clarifying where the method could be further improved in the next stage.

Final Summary Section (End of Course)

- **Overall achievements:**

1) Developed a Gradio based interactive system with 3 different tabs: VGG NST, Stable Diffusion Img2Img and Person-aware Local Edit.
2) Used two-stage hybrid pipeline (VGG19 draft + SD v1.5 refinement) and introduced fashn-human-parser to make semantic mask.
3) Allowed for 3 types of local edit that could be done using a mask: palette recolouring, pattern print transfer, background replacement (solid / style / diffused).
4) Performed comparative experiments between pure VGG, pure SD and Hybrid modes and moved the UI from PyQt5 to Gradio and gave the status report while training.

- **Major difficulties overcome:**

1) The use of torch. no_grad() during feature extraction and freezing of the parameters in VGG19 reduce VRAM usage and prevent UI blocking.
2) Paved the way for bridging the gap between tensor and PIL format between VGG and SD stages with a custom created tensor_to_image() conversion pipeline.
3) Accurate ROI heuristics such as rectangles have been replaced with human parsing masks per pixel, to ensure robust localization of garment.
4) Get progress feedback synchronized among different models: implement a weighted callback system.

- **Skills learned:**

1) Technical: Custom PyTorch pipelines and Diffusers inference, UI development with Gradio, and Mask based compositing of images.
2) Research: Controlled comparative experiments, bad-case documentation and scientific writing in IEEE format.

- **Outcomes (report, model, prototype, presentation):**

1) Final Report: Report with system architecture, mathematical formulations, IEEE citations and experimental analysis.
2) Prototype: A Gradio web app that can be deployed with run_local. ps1 and run_lan. ps1 for uploading, local editing based on the mask and two-stage generation.
3) Pipeline: Reproducible multi-model backend (VGG19 + SD v1.5 + human parser) and config management and model download scripts.
4) Presentation: Group oral presentation including live demonstrations, comparative cases and application scenarios.